

KYLE NERO A. SANTOS

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PROFESSIONAL SUMMARY

Electronics Engineering graduate specializing in embedded systems, computer vision, and software development. Experienced in designing hardware-software solutions using ESP32 and YOLO-based object detection, alongside developing cross-platform applications and enterprise workflow automation.

EDUCATION

Bachelor of Science in Electronics Engineering 2022 - Present
Marikina Polytechnic College
Capstone: Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use

INTERNSHIP

First Option Electronic System Phils., Inc. June 2025 - Aug 2025

- Managed end-to-end surveillance operations from designing comprehensive CCTV layouts to overseeing control room personnel, enhancing overall site security and operational efficiency.

TECHNICAL SKILLS

Programming: Python, OpenCV, YOLOv11, C/C++, Arduino/ESP32 firmware logic
Computer Vision and AI: object detection, dataset annotation, model training, AI Automation
API and Data Handling: API concepts, JSON, integrating AI/API-backed features, Database Management
Tools and Simulation: VS Code, Antigravity IDE, Claude Code, Arduino IDE, Excel, MS Office
ITSM Automation: ServiceNow Basic Fundamentals, ServiceNow Catalog (Including and not limited to Creation of Catalog Item, Basic Flow Designer, And System Administration/Navigation)

ENGINEERING PROJECT EXPERIENCE

Capstone Project Lead and Full-System Developer | MICROPLASTIC DETECTOR April 2026

- Engineered a Raspberry Pi 5 optical detection prototype, overcoming the time and cost limits of manual rock salt microscopy.
- Implemented a YOLOv11 and OpenCV detection workflow, achieving 0.826 precision, 0.733 recall, and 0.814 mAP@50.
- Developed an end-to-end operator interface, automating the generation of microplastic contamination and particle count reports.
- Executed controlled multi-trial testing across diverse rock salt samples, validating prototype performance under varied conditions.

ADDITIONAL PROJECTS

App Projects | HerCare and KKB Snap Product and Frontend Development

- Developed interactive mobile product flows for a women's health companion app (HerCare) using React, TypeScript, and Vite, delivering a seamless, user-focused interface.
- Implemented Capacitor Android and a Supabase backend for a bill-splitting application (KKB Snap), enabling cross-platform deployment and real-time expense data sharing.

CERTIFICATIONS

ServiceNow Administration Fundamentals In Progress

- Pursuing comprehensive platform administration training, preparing for the official Certified System Administrator (CSA) credential.
- Completed foundational system administrator microlearning modules, establishing a practical baseline for enterprise workflow management.

AFFILIATIONS

- Institute of Electronics Engineers of the Philippines, Student Member